11 The average orbital speed v of an object in the Solar System is defined by the equation

$$v = \frac{2\pi r}{T}$$

(a) Define the terms r and T in this equation.

r is .....

T is .....

[2]

(b) The average orbital speed of Saturn is 9600 m / s.

(i) Explain why the orbital speed is an average value.

.....

.....

..... [2]

(ii) Saturn is 1.4×10^{12} m from the Sun.

Show that one Saturn year is approximately equal to 30 Earth years.

[4]

[Total: 8]

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